

DSC 102

Systems for Scalable Analytics

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Final Prep-up Material: Exercises

Parallelism, Computational Cost, Map-Reduce

Exercise 1

Assume you parallelize a data-intensive program across multiple cores on a single machine. 30% of the program is inherently sequential, while the rest can yield perfect linear speedup for any number of cores.

The single-core runtime of the program is 90min.

1. What is the runtime (in minutes) of parallel 9-core execution?

Answer: The time consumed in sequential execution: $90' * (30/100) = 27'$.

Remainder to be parallelized: $90' - 27' = 63'$.

Over 9 cores, perfect linear speedup means execution will take $63' / 9 = 7'$.

Total execution time: $27' + 7' = 34'$.

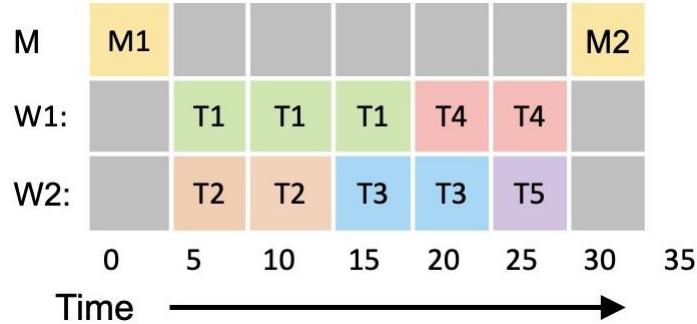
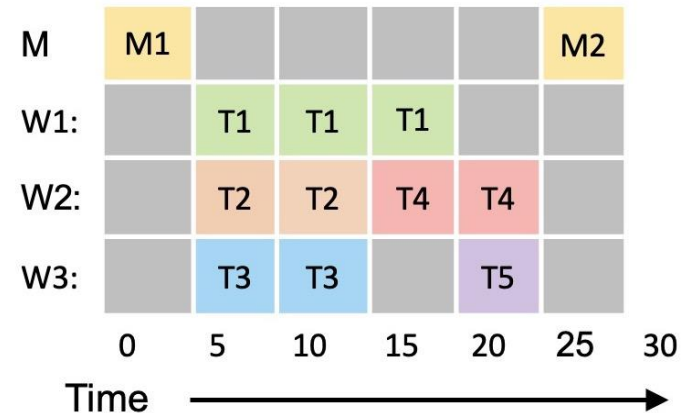
2. What is the overall speedup factor for the program? If needed, use up to 1 decimal digit to approximate your answer.

Answer: $90' / 34' = 2.6$

Exercise 2

Consider this workload with 5 tasks T1 – T5 being run in a task-parallel manner on a system with 3 workers and a manager node. Task T5 depends on T1. When a task starts, it runs uninterrupted until it completes on the same worker. With these rules, it takes 30 time units for the workload to complete on 3 workers.

What is the lowest possible task-parallel completion time of this workload given only 2 workers W1 and W2?



Answer: With 2 workers, any 2 the T1-T4 tasks can start running in parallel. Let us assume first we have T1 run on W1 and T2 run on W2.

T2 will be done faster than T1. T3 or T4 can then run on W2.

Let us assume T3 follows T2. This means T4 or T5 can run after T1 is complete on W1.

However, to achieve lowest completion time, T4 must follow T1 on W1, and T5 must start on W2 after T3 is complete. (Otherwise, T4 will be starting on W2 when T5 is done on W1.)

The workload will complete at the lowest of 35 time units.

Exercise 3: Compute Cost

Use case of amplifying the accuracy of a linear model with a multi-step workflow.

You are given a large training dataset with schema $D (Y, X_1, X_2, \dots, X_d)$. The target Y and all features are numeric and stored as float64.

The number of examples is n . The dataset is laid out in row store format on disk (no compression) on your single machine. No batching of file scans or caching of data is performed.

Step A. You build linear models using BGD. You try 10 different learning rates and train all models for 3 epochs each.

Step B. You read the data and append quadratic (second order) pairwise feature interactions. You write out the output file to disk for future use. All features are still stored as float64.

Given: ($n = 500$ million, $d = 2$)

First, let us process information: Each row has $d+1$ numbers $\Rightarrow (d+1)*64$ bits $\Rightarrow (d+1)*8$ bytes

For the given $d=2 \Rightarrow$ The dataset has $d+1 = 2+1 = 3$ columns, and each row is $3*8=24$ bytes
 $n = 500,000,000$ or $n = 5e8$

A indicates there are 10 learning rates, so there is a total of 10 different models.

A also indicates 3 epochs for each model. This is BGD, so each epoch is a full dataset scan.

B specifies quadratic pairwise feature interactions, so we need to consider as a new feature each one of all possible combinations $X_1X_1, X_1X_2, \dots, X_1X_D, X_2X_3, \dots, X_2X_D, \dots, X_DX_D$.

1. What is the size in GB of the file D on disk?

Answer: The size is (row size)*(# rows) = $24 * 5e8 = 120e8 = 12e9$ bytes = **12** GB

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Given: ($n = 500$ million, $d = 2$)

2. What is the total I/O cost in GB of step A for reading data files?

Answer: For Step A, we need to read the dataset $10 \times 3 = 30$ times $\Rightarrow 12 \text{ GB} \times 30 = 360 \text{ GB}$.

3. What is the size in GB of the file written out in step B?

Answer: For $d=2$: Interactions are among the pair: (x_1, x_2) . This means we have to consider the interaction terms x_1x_1, x_1x_2, x_2x_2 . We need a separate column for each one of these interactions; therefore 3 more columns are needed. The updated dataset will have 6 columns. Each row will now consume $6 \times 8 = 48$ bytes.

Total output will be $(\text{row size}) \times (\# \text{ rows}) = 48 \times 5e8 = 240e8 \text{ bytes} = 24 \text{ GB}$.

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Given: ($n = 500$ million, $d = 2$)

4. What is the degree of parallelism of this entire 2-step workflow if you were to run it in a task-parallel manner?

Answer: A different worker can be used to run all the epochs for each learning rate. So for the 10 learning rates in step A we can run the step on 10 workers simultaneously.

Step B can be performed independently from A, hence an additional worker can be used for it. Therefore, the total number of $10+1 = 11$ workers that can be used simultaneously is the degree of parallelism for this use case.

Exercise 4: Workflows

Assume you need to run a classification workflow with 3 alternative feature engineering (FE) approaches. On each feature set, you apply a Random Forest (RF) model with 2 different hyperparameter combinations for each FE approach.

1. How many workers are necessary to achieve lowest-possible completion time with task parallelism no matter the tasks' runtimes?

Answer: Lowest completion time can be achieved when we take maximum advantage of parallelism. Therefore, the question just asks what this workflow's degree of parallelism is. To compute this, we need to calculate the number of different workflow variations that can be each run independently on a different worker. The number of variations are the possible combinations across all alternative FE approaches and RF models, which is $3 \times 2 = 6$. Therefore, one needs $3 \times 2 = 6$ workers.

2. Assume the task runtimes are known as follows:

- 30, 35, and 40 units for the FE approaches.
- For the 2 different hyperparameter combinations in each feature set: 20 and 30 units to build the RF models.

What is the upper bound on the lowest-possible completion time for this workflow?

Answer: We need an upper bound on the cost of the longest path. The longest path would be at most 40 units for the FE approach, and at most 30 units for the RF models. So the longest path upper bound will be $40 + 30 = 70$ time units.

Exercise 4: Workflows

Assume you need to run a classification workflow with 3 alternative feature engineering (FE) approaches. On each feature set, you apply a Random Forest (RF) model with 2 different hyperparameter combinations for each FE approach.

3. What is the speedup when hitting the upper bound vs. using just 1 worker with no idle times for your whole workflow? If needed, use up to 1 decimal digit to approximate your answer.

Answer: For the speedup we need the ratio $s = t_1 / t_n$,

where t_1 =[Completion time given 1 worker] and t_n =[Completion time given n workers].

The completion time t_n will be the slowest possible combination of FE approaches and RF models. The answer about the longest path upper bound in question 2 tells us exactly this completion time value that is $t_n = 70$.

For the total 1-worker time we sum up the times needed:

[time for all FE approaches] + [time to run both RF models for each of the 3 FE approaches] =
[30+35+40] + [3*(20+30)] = 105 + 150 = 255 = t_1

In conclusion, the requested speedup is $255/70 = 3.6$ (1 decimal digit approximation.)

Exercise 5: Compute Cost

Assume you are given a large dataset file for ML training that is of size 120 GB. What is the lowest possible I/O cost (in GB) of each of the following feature engineering operations? Ignore final output write costs and any potential gains due to caching.

1. Quadratic (order2) feature interactions.

Answer: 120 GB (only a single file scan is needed).

2. Binning a numeric feature into 10 given intervals.

Answer: 120 GB (only a single file scan is needed).

3. One-hot encoding of a categorical feature (assume feature's domain has only 5000 unique values).

Answer: 120 GB (only a single file scan is needed).

4. Whitening a numeric feature by normalizing the values.

Answer: 240 GB (one file scan is needed to compute mean and stdev; a second file scan is needed to whiten feature values).

Exercise 6: Map-Reduce

You have the dataset
 $A = [1, 5, 5, 9, 15, 20, 30]$
and you want to program map-reduce code
to compute the average of the numbers in A
on a 3-worker system.

For the reduce phase, your code for the
Manager node only adds up the partial
results of the Worker nodes to produce the
desired outcome.

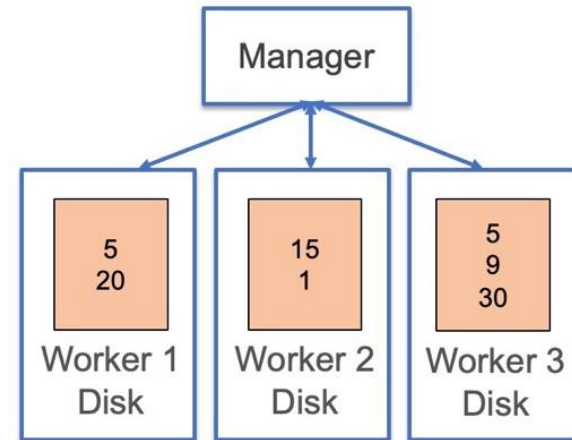
Given the above information, and that

- the data are sharded randomly across the workers drives as shown in the figure, and
 - the Manager node passes as input to each Worker node the population size of A
- you want to use the correct algebraic implementation in your code to calculate the average of A .
Answer the following questions. If needed, use 4 decimal digits to approximate your answer.

1. What is the average of A ?

Answer: A is a set with population size 7. The average value is

$$(1+5+5+9+15+20+30) / 7 = 12.1428$$



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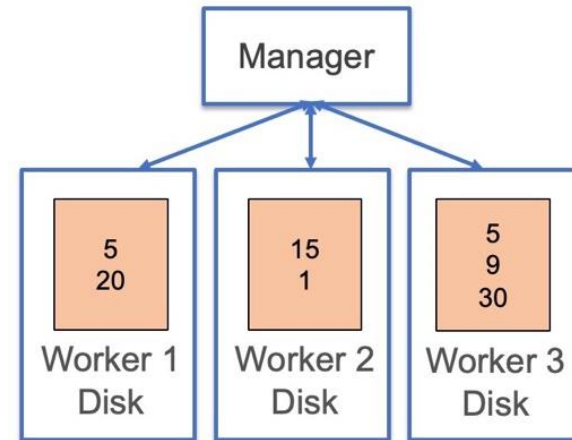
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Answer the following questions. If needed, use 4 decimal digits to approximate your answer.

2. What is the partial result that Worker 1 should return to the Manager node?

Answer: Worker 1 should return the weighted average of its data shard:

$$2/7 * (5+20)/2 = 3.5714$$



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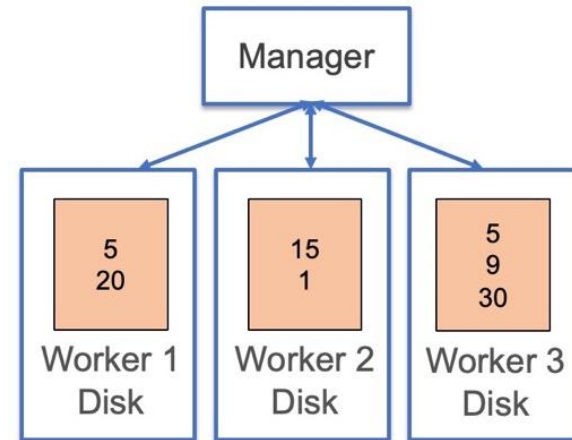
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you want to use the correct algebraic implementation in your code to calculate the average of A .
Answer the following questions. If needed, use 4 decimal digits to approximate your answer.

3. What is the partial result that Worker 2 should return to the Manager node?

Answer: Worker 2 should return the weighted average of its data shard:

$$2/7 * (15+1)/2 = 2.2857$$



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and you want to program map-reduce code to compute the average of the numbers in A on a 3-worker system.

For the reduce phase, your code for the Manager node only adds up the partial results of the Worker nodes to produce the desired outcome.

Given the above information, and that

- the data are sharded randomly across the workers drives as shown in the figure, and
- the Manager node passes as input to each Worker node the population size of A

you want to use the correct algebraic implementation in your code to calculate the average of A . Answer the following questions. If needed, use 4 decimal digits to approximate your answer.

4. What is the partial result that Worker 3 should return to the Manager node?

Answer: Worker 3 should return the weighted average of its data shard:

$$3/7 * (5+9+30)/3 = 6.2857$$

Note 1: Observe: The sum of the partial results correctly gives the sought average of the set A .

Note 2: If it were specified that in the reduce phase the Manager node can perform more elaborate computations than just add partial results, an alternative solution would be for each Worker to return just the subset average. The Manager node should then weigh each Worker's partial result by the ratio of the Worker's subset population over the A population.

